

Power-Saving Relay Driver

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In many circuits, the switching action is performed by a relay, which in turn activates an external load. The power consumed by the relay may be unsuitable for battery-powered applications. Here is a simple solution using some inexpensive components to considerably save power.

Circuit and working

Fig. 1 shows circuit diagram of the power-saving relay driver where resistor R1 and transistor T1 form a stand-

ard relay driver circuit.

Once the relay is energised, its pole is pulled in to make contact with the N/O side, and it holds in that position with typically 75 per cent of its nominal-rated voltage. Power consumed by a relay coil during this holding time equals V^2/R , where R is resistance of the relay coil and V is the voltage. Here resistors R2 and R3, transistor T2 and capacitor C1 lower the power consumption after actuation by applying less than the normal operating power.

Initially, when power is applied, capacitor C1 momentarily shorts resistor R2 and allows full voltage across the relay to pull the pole contact, and then slowly the current through the capacitor drops.

In the meantime, resistor R2 takes care of the current, ensuring it is just sufficient to hold the relay. The constant current mechanism formed by transistor T2 and resistors R2 and R3 effectively drives the relay at very less power.

Following calculations will help us understand how additional circuitry around relay driver transistor T1 saves power. Relay used here is a 12V, 400-ohm sugar-cube type. You can calculate the power saved as shown below:

Nominal current required for the relay (I) = $12V/400\text{ ohm} = 30\text{mA}$

Power consumed by the relay = $I^2R = 0.03A \times 400\text{ ohm} = 360\text{mW}$

After introduction of the circuit:

The current through the coil (I) = $V_{BE}/R_2 = 0.6V/47\text{ ohm} = 12\text{mA}$

Power consumed by the overall circuit = $V \times I = 12 \times 0.012 = 144\text{mW}$

Power saved = $360\text{mW} -$



Test Points

Test point	Details
TP0	0V
TP1	12V
TP2	High when switch S1 is pressed or high through connector CON2

PARTS LIST

Semiconductors:	
D1	- 1N4001 rectifier diode
T1, T2	- BC547 npn transistor
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1, R3	- 47-kilo-ohm
R2	- 47-ohm
Capacitor:	
C1	- 220 μ F, 25V electrolytic
Miscellaneous:	
CON1	- 2-pin terminal connector
CON2	- 2-pin connector
CON3	- 3-pin connector
RL1	- 12V, 400-ohm 1C/O relay
S1	- Tactile switch

144mW = 216mW

So, we conclude that considerable power can be saved using the additional circuitry.

This makes it fairly simple for anyone to re-design a relay driver to reduce its power consumption without the use of any expensive components.

Construction and testing

An actual-size, single-side PCB for the power-saving relay driver is shown in Fig. 2 and its component layout in Fig. 3.

Switch S1 is used to test the relay driver circuit. You can connect the output of a control circuit, such as a microcontroller, to CON2 for controlling the relay circuit.

CON3 helps in connecting to the electrical load. You can connect the load between N/O and pole contacts or N/C and pole contacts.

Before connecting the load to CON3, verify the test point voltages given in the table. You may reduce the value of R1 as per your requirement. ●

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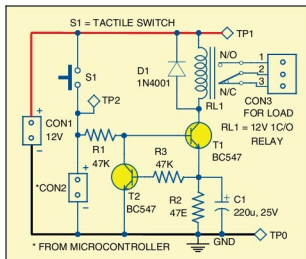


Fig. 1: Circuit diagram of the power-saving relay driver

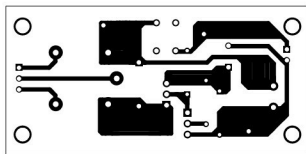


Fig. 2: An actual-size, single-side PCB for the power-saving relay driver

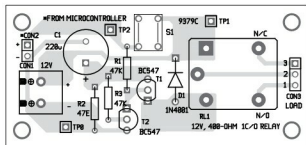


Fig. 3: Component layout for the PCB